

Tutorial - 9

(67)

Name : Varadray Satish Patil

16/3/2026

Class : CSSE-B

Roll no : 12412485

Roll no : 65

Problem Statement : Evaluate software risks in a project and propose suitable risk management strategies.

Solution :

Introduction :

In any software project, risks are uncertain events that can negatively impact costs, schedule, and quality. Effective risk management and configuration management ensure project stability, reliability, and success.

What is Software Risk?

A software risk is a potential problem that may occur during development and deployment.

Types of risks

Technical risks

Use of new or untested technologies

Integration issues

Performance bottlenecks

Project risks

Budget overruns

Delays in delivery

Resource shortages

Business risks

- Market changes
- Customer dissatisfaction
- Legal or compliance issues.

3. Risk Evaluation Process

Step 1: Risk identification

→ Identify possible risks using:

- Brain storming
- Expert consultation
- Past project analysis.

→ Example risks:

- Requirement changes
- Developer attrition
- Security vulnerabilities.

Step 2: Risk analysis

→ Assess each risk based on:

- Probability (P): likelihood of occurrence
- Impact (I): severity of consequences

→ Risk Exposure formula:

$$RE = P \times I.$$

Step 3: Risk Prioritization

- High RE → Immediate attention
- Medium RE → monitor regularly
- Low RE → accept or ignore.

4. Risk management strategies

i) Risk avoidance

- Change project plan to eliminate risk
- Example: avoid using unstable technology

ii) Risk mitigation

- Reduce probability or impact

Example: Training developers.

Risk transfer

Shift risk to third party

Example: Outsourcing modules.

Risk acceptance

Accept risk with contingency plans.

Risk management plan

Risk list

Risk owner

Mitigation actions

Monitoring strategy

Sample Table:

Risk	Probability	Impact	Strategy
Requirement changes	high	high	Agile methodology
Developer leaves	Medium	high	Backup resources
Security breach	low	very high	Security testing

Configuration management (CM)

Definition: CM ensures that software changes are tracked, controlled, and documented.

Objectives of configuration management

Maintain consistency

Control changes

Enable traceability

Prevent errors.

Key components of CM.

Configuration Identification

Identify items (code, documents, requirements)

Version Control
Maintain different versions of software

Tools: git, SVN

Change management.
Process of handling change requests:

Submit request

Analyze impact

Approve / reject

Implement

Configuration status accounting

Track status of changes.

Configuration Audits

Ensure correctness and compliance

1. CM strategies for projects

Use version control systems.

Maintain code history

Enable collaboration

2) Branching strategy

Main branch (stable)

Feature branches (development).

3) Automated builds & CI/CD

Continuous integration ensures early detection of errors.

4) Access control

Restrict unauthorized changes.

5) Documentation management

Maintain updated documentation

6) Best practices

Identify risks early.

Continuously monitor risks
Use automation tools
Maintain proper documentation
Conduct regular audits.

Conclusion

Effective risk evaluation and management, combined with strong configuration management practices, ensures:

Reduced project failure
Improved software quality
Better team coordination
Successful project delivery.

Scm